

Cybersecurity Lab Report

Vulnerability Scanning with Nessus & OpenVAS

Target: Metasploitable2 (192.168.255.128) • Attacker: Kali Linux • Date: May 30, 2026

1. Introduction

This report documents the second phase of the cybersecurity lab, extending the initial Nmap reconnaissance with active vulnerability scanning using two industry-standard tools: Nessus Essentials (by Tenable) and OpenVAS/GVM (Greenbone Vulnerability Manager). Both scanners were installed and operated directly on the Kali Linux attacker machine, targeting the Metasploitable2 Docker container at 192.168.255.128.

The objectives of this phase are:

- Install and configure Nessus Essentials on Kali Linux and verify the service is running.
- Run an Advanced Scan against the target and review the vulnerability findings by severity.
- Install and configure OpenVAS/GVM (Greenbone 25.04) on Kali Linux and start all required GVM services.
- Create a scan target in OpenVAS and execute the first vulnerability scan (Project #1 Scan).
- Analyze the OpenVAS results through Tasks, Reports, Results, Vulnerabilities, and Dashboards views.

Unlike Nmap, which only performs port and service enumeration, Nessus and OpenVAS actively probe each discovered service against a database of known vulnerabilities (CVEs), misconfigurations, outdated software, and default credentials, producing prioritized, actionable reports for remediation.

2. Nessus Essentials — Installation and Scan

2.1 — Installing Nessus on Kali Linux

Nessus Essentials version 10.12.0 was downloaded as a Debian package (Nessus-10.12.0-ubuntu1604_amd64.deb) and installed using `sudo dpkg -i`. The installer ran a full FIPS (Federal Information Processing Standard) cryptographic self-test, validating all required cipher and hash algorithms before completing the installation.

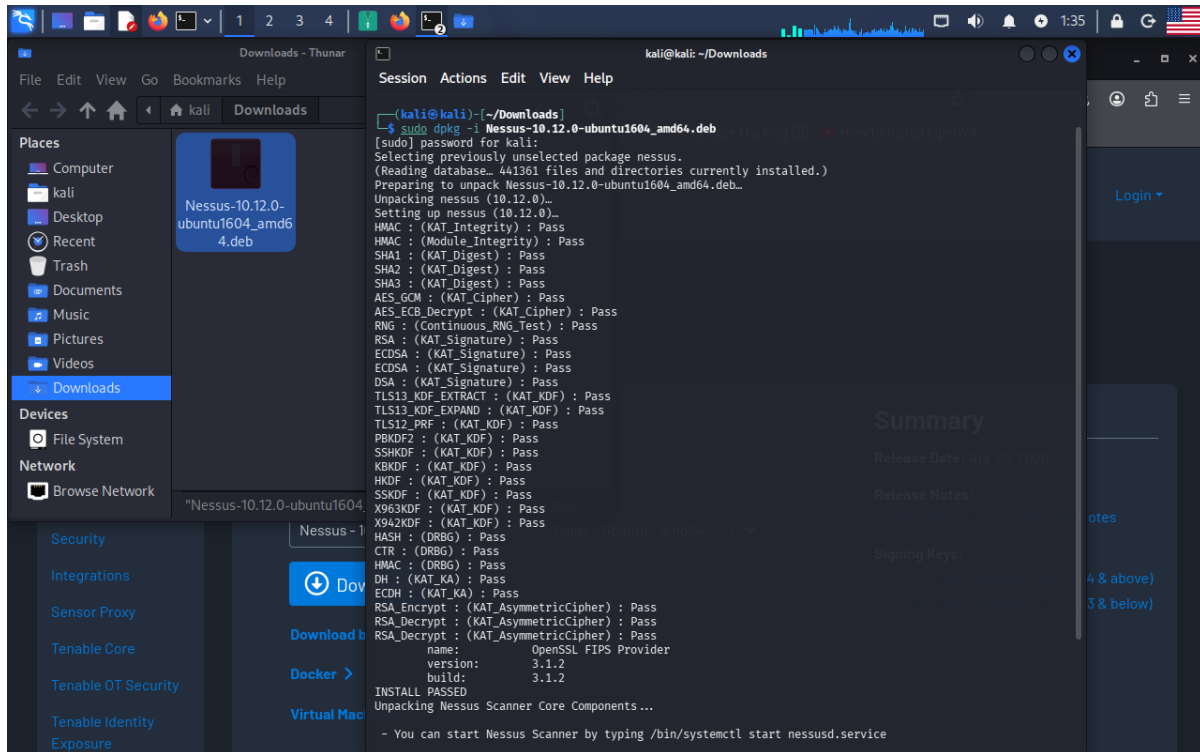


Figure 1 — Kali terminal: `sudo dpkg -i Nessus-10.12.0-ubuntu1604_amd64.deb` installs Nessus 10.12.0. The output shows the full KAT (Known Answer Test) cryptographic validation suite passing for HMAC, SHA-1/2/3, AES-GCM, AES-ECB, RNG, RSA, ECDSA, DSA, TLS, PBKDF2, SSHKDF, KHKDF, HKDF, SSKDF, X963KDF, HASH, CTR, DH, ECDH, and RSA asymmetric cipher functions. OpenSSL FIPS Provider version 3.1.2 is confirmed. Installation concludes with the message: "You can start Nessus Scanner by typing `/bin/systemctl start nessusd.service`".

2.2 — Starting the Nessus Service

After installation, the Nessus daemon was started with `/bin/systemctl start nessusd.service` and its status was verified. An initial typo in the service name ("nessusd.servive") produced a "Unit not found" error, after which the correct command confirmed the service as active and running with PID 105579, consuming 178.8 MB of memory. The Nessus web interface was then accessible at localhost:8834, showing the "My Scans" page ready for use.

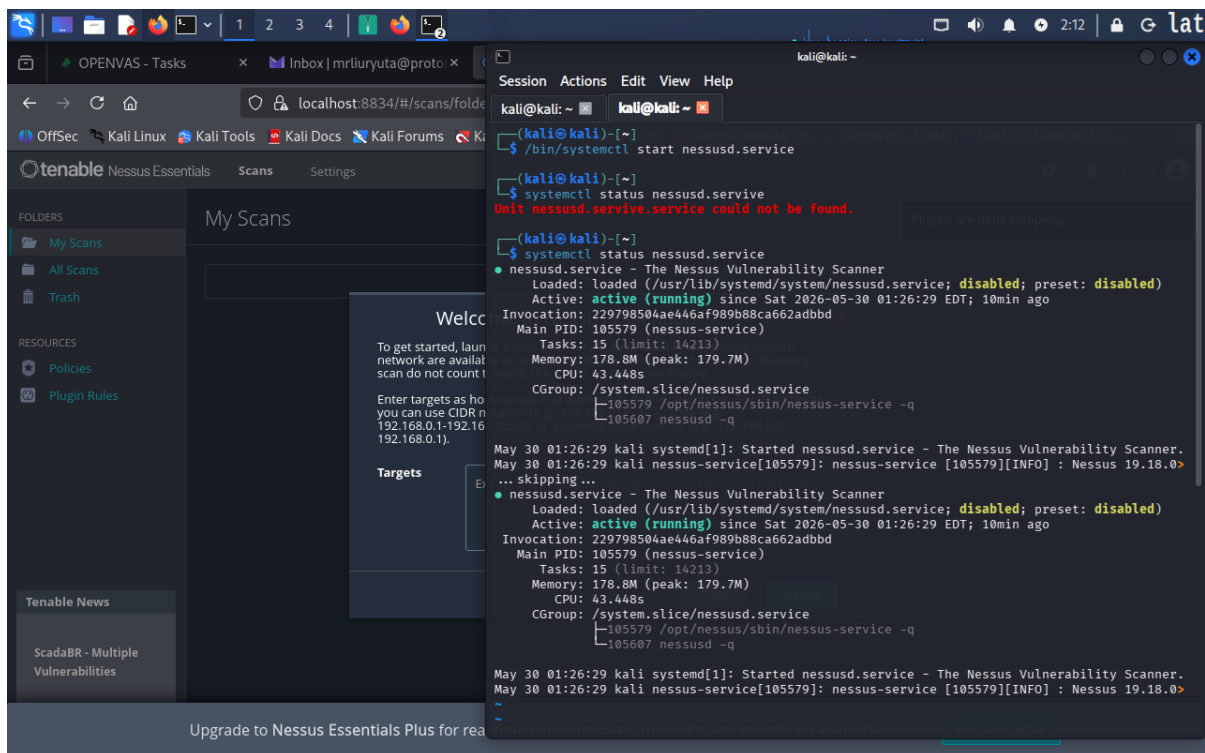


Figure 2 — Kali terminal overlaid on the Nessus Essentials web interface (localhost:8834). Terminal shows: (1) failed attempt with typo "nessusd.servive" — "Unit could not be found"; (2) correct `systemctl status nessusd.service` output confirming the service is active (running) since Sat 2026-05-30 01:26:29 EDT, PID 105579, memory 178.8 MB, 15 tasks; (3) systemd journal confirming "Started nessusd.service — The Nessus Vulnerability Scanner" and Nessus 19.18.0 INFO log. The browser behind shows Nessus loading with the "Plugins are still Compiling" notice.

2.3 — Creating and Launching the Nessus Scan (Analysis#1 — Running)

A new vulnerability scan named "Analysis#1" was created in Nessus Essentials using the Advanced Scan policy and configured to target 192.168.255.128. The scan was launched on demand and the My Scans page confirmed it was active.

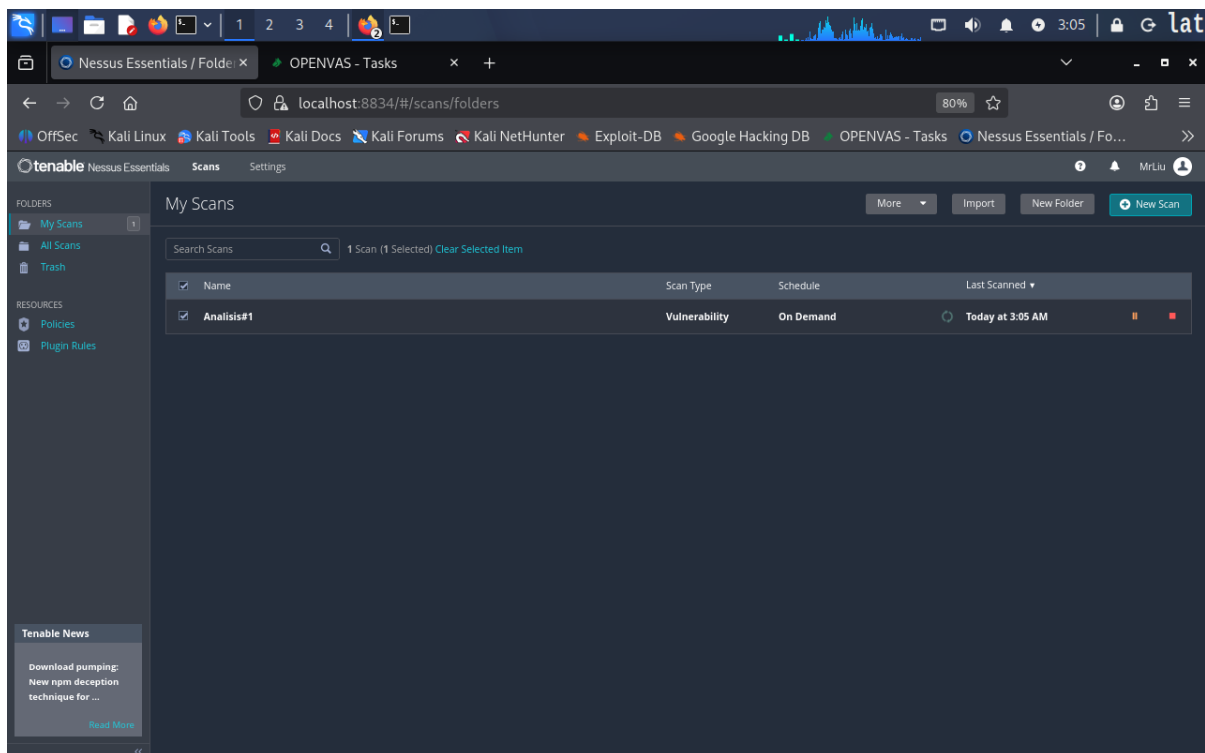


Figure 3 — Nessus Essentials web interface — My Scans view (localhost:8834/#/scans/folders). The scan "Analysis#1" is listed with Scan Type: Vulnerability, Schedule: On Demand, and Last Scanned: Today at 3:05 AM. The scan is selected (checkbox active) and ready to be opened. The sidebar shows My Scans (1), All Scans, Trash, Policies, and Plugin Rules under Resources.

2.4 — Nessus Scan in Progress (Hosts Tab)

Opening the Analysis#1 scan while it was still running revealed the Hosts tab with a live vulnerability count bar for the target. At this point only 22 unique vulnerabilities had been detected with the scan at 7% completion, confirming the scan was actively enumerating services.

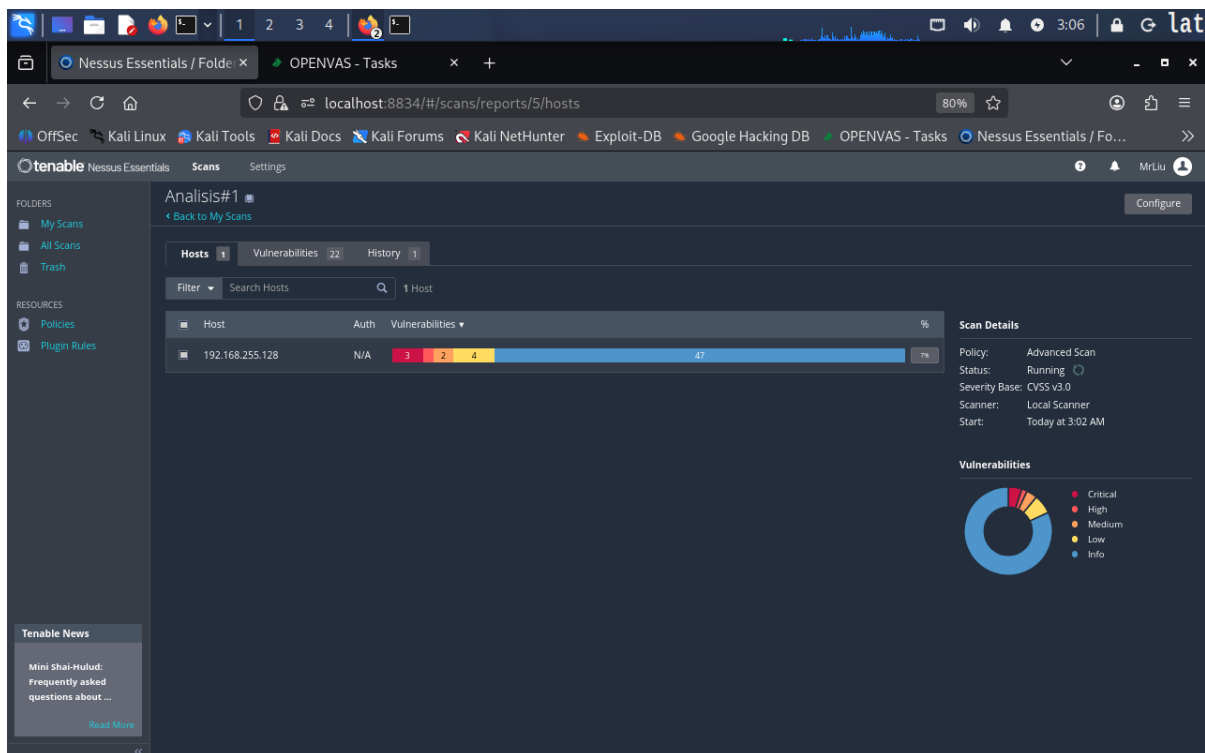


Figure 4 — Nessus Essentials — Analysis#1 scan detail, Hosts tab (localhost:8834/#/scans/reports/5/hosts). Status: Running (spinning indicator). Target host 192.168.255.128, Auth: N/A (not yet attempted). Vulnerability bar shows 3 Critical (red), 2 High (orange), 4 Medium (yellow), 47 Info (blue) at 7% progress. Scan Details panel: Policy: Advanced Scan, Status: Running, Severity Base: CVSS v3.0, Scanner: Local Scanner, Start: Today at 3:02 AM. Vulnerability donut chart visible with Critical, High, Medium, Low, Info categories. Vulnerabilities tab shows count 22 (in progress).

2.5 — Nessus Scan Completed (Hosts Tab with Final Results)

Once the scan completed, the Hosts tab updated to show the final vulnerability tally. The Auth column changed to "Fail" confirming unauthenticated scanning only, and the full vulnerability bar revealed the complete distribution across all severity levels.

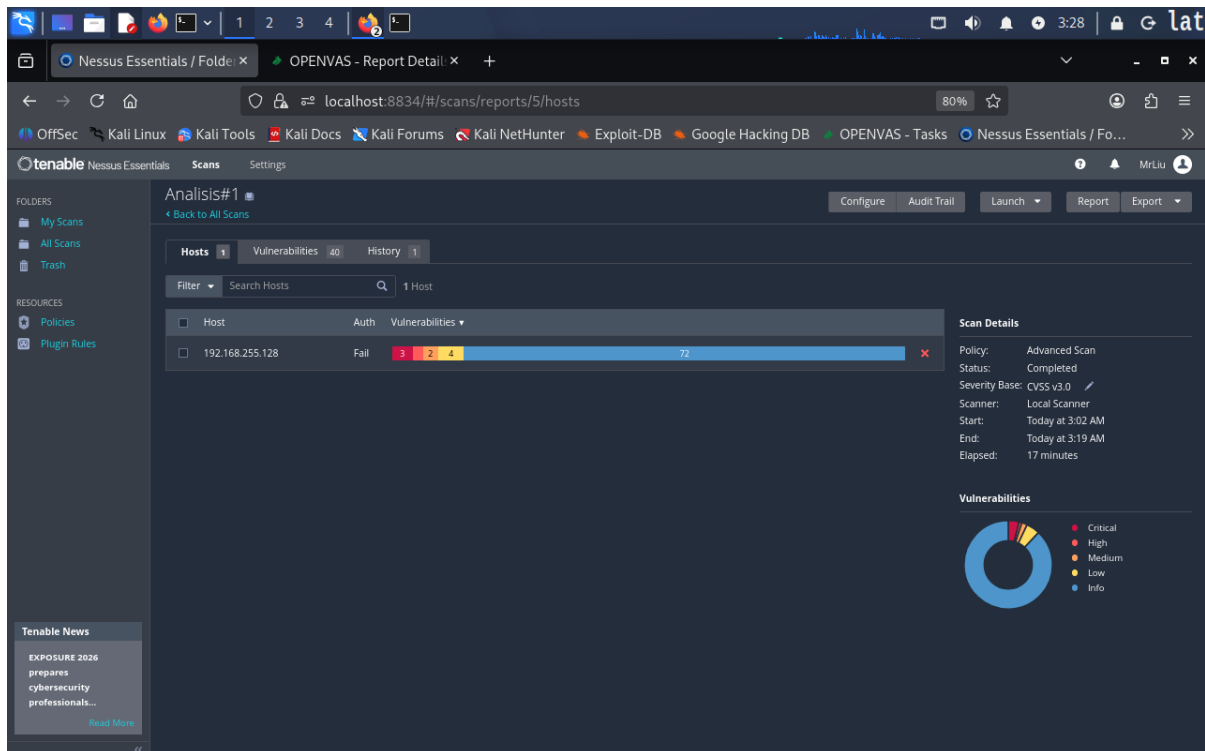


Figure 5 — Nessus Essentials — Analysis#1 scan completed, Hosts tab (`localhost:8834/#/scans/reports/5/hosts`). Target 192.168.255.128, Auth: Fail. Vulnerability bar: 3 Critical (red), 2 High (orange), 4 Medium (yellow), 72 Info (blue). Scan Details: Policy: Advanced Scan, Status: Completed, Severity Base: CVSS v3.0, Scanner: Local Scanner, Start: Today at 3:02 AM, End: Today at 3:19 AM, Elapsed: 17 minutes. Full action toolbar visible (Configure, Audit Trail, Launch, Report, Export). Donut chart shows the same severity breakdown with Info dominating.

2.6 — Nessus Vulnerabilities Tab (Full Plugin List)

The Vulnerabilities tab presented the complete list of 40 unique plugin findings sorted by severity. This view shows each plugin grouped by its severity badge (Critical, High, Mixed, Low, Info) alongside CVSS score, VPR, EPSS probability, vulnerability name, Nessus family, and instance count.

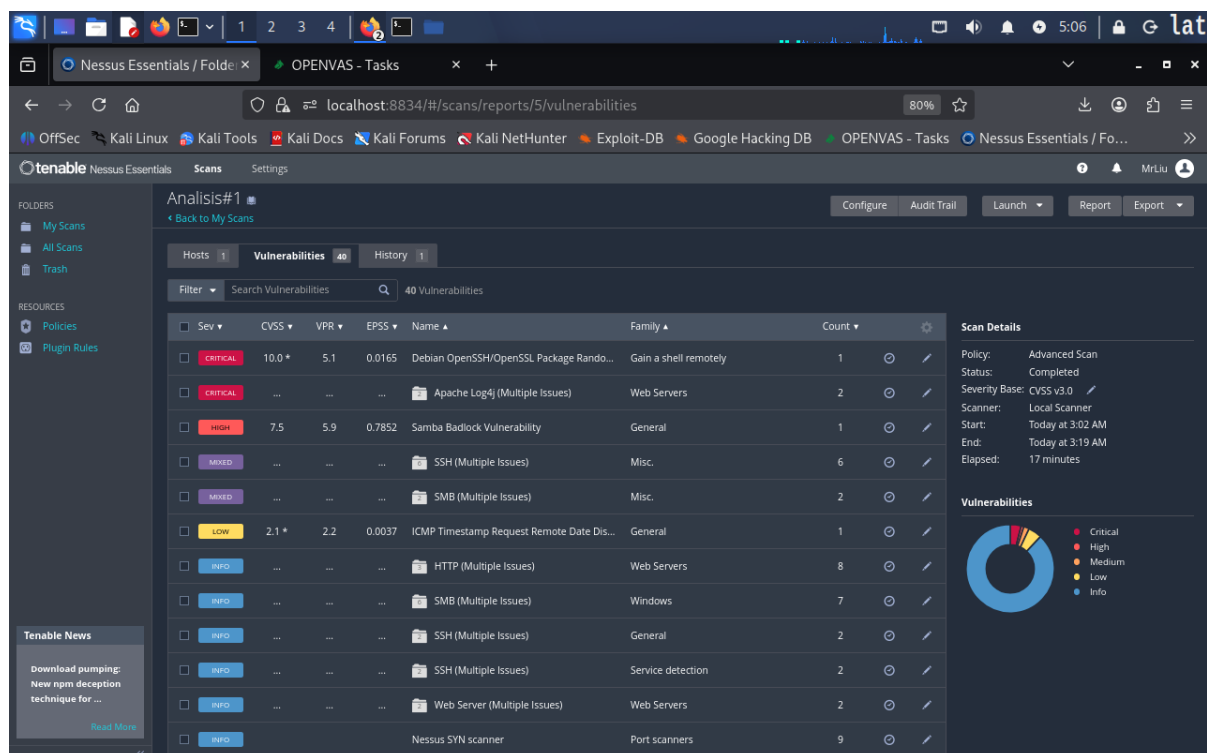


Figure 6 — Nessus Essentials — Analysis#1, Vulnerabilities tab (localhost:8834/#/scans/reports/5/vulnerabilities) showing 40 Vulnerabilities. Columns: Severity, CVSS, VPR, EPSS, Name, Family, Count. Top findings: (1) CRITICAL — Debian OpenSSH/OpenSSL Package Randomness Weakness, CVSS 10.0*, VPR 5.1, EPSS 0.0165, Family: Gain a shell remotely, Count 1; (2) CRITICAL — Apache Log4j (Multiple Issues), Family: Web Servers, Count 2; (3) HIGH — Samba Badlock Vulnerability, CVSS 7.5, VPR 5.9, EPSS 0.7852, Family: General, Count 1; (4) MIXED — SSH (Multiple Issues), Misc., Count 6; (5) MIXED — SMB (Multiple Issues), Misc., Count 2; (6) LOW — ICMP Timestamp Request Remote Date Disclosure, CVSS 2.1*, VPR 2.2, EPSS 0.0037, General, Count 1; followed by INFO rows for HTTP (8), SMB/Windows (7), SSH/General (2), SSH/Service detection (2), Web Server (2), Nessus SYN scanner (9).

3. OpenVAS / GVM — Installation and Scan

3.1 — GVM Setup and Service Startup

OpenVAS was installed via the Kali Linux package manager as part of the Greenbone Vulnerability Manager (GVM) suite version 25.04.0. After installation, `gvm-setup` was run to initialize the PostgreSQL database, download SCAP and CERT data feeds, and verify all component services. The setup script confirmed all 9 checks passing before marking the installation as complete.

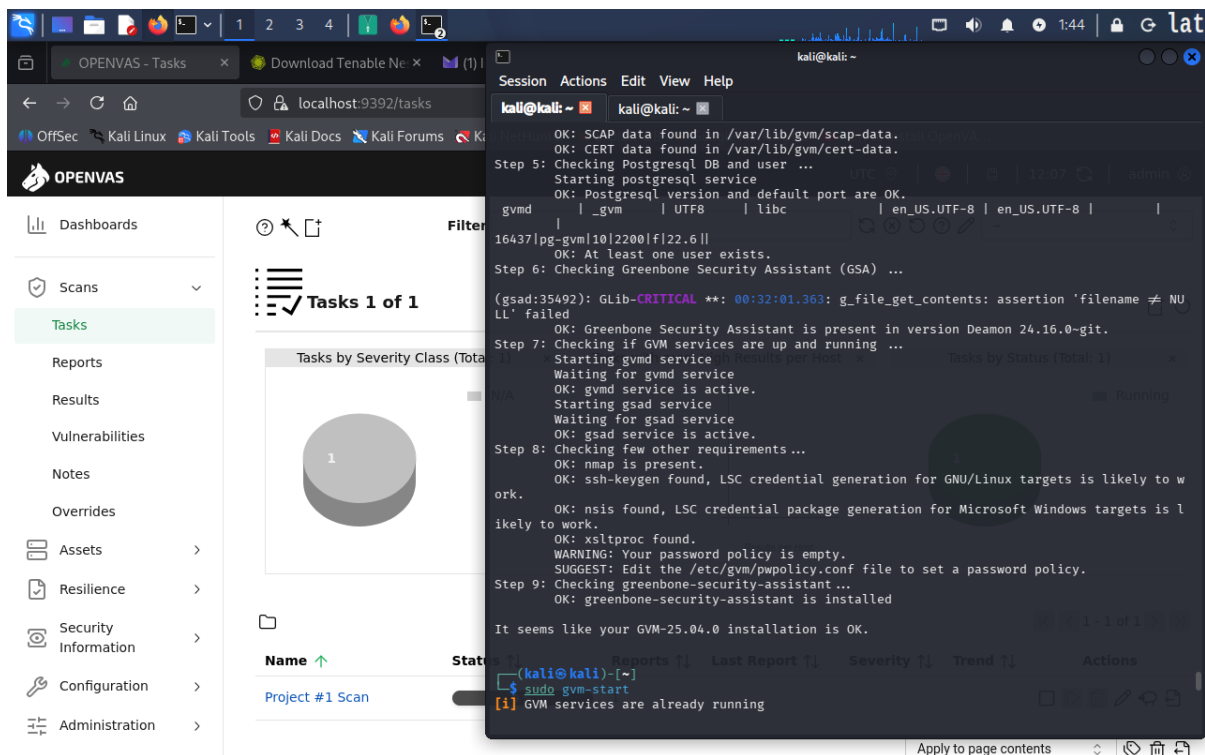


Figure 7 — Kali terminal showing the tail end of `gvm-setup` output (overlaid on the OpenVAS Tasks web interface). Setup output confirms: OK: SCAP data found in `/var/lib/gvm/scap-data`; OK: CERT data found in `/var/lib/gvm/cert-data`; Step 5: PostgreSQL DB initialized, version and port OK; `gvmd` database (`pg-gvm 10`, `gvmd 22.6`); OK: at least one user exists; Step 6: GSA (Greenbone Security Assistant) present in version Deamon 24.16.0-git (with a GLib CRITICAL warning for `g_file_get_contents`); Step 7: `gvmd` service active, `gsad` service active; Step 8: `nmap` present, `ssh-keygen` found (LSC credential generation likely to work), `nsis` found, `xsltproc` found; WARNING: password policy is empty; Step 9: `greenbone-security-assistant` installed. Final message: "GVM-25.04.0 installation is OK." Then `sudo gvm-start` is run and "[1] GVM services are already running" is displayed.

3.2 — OpenVAS Target Configuration (PROJECT #1)

Before running a scan, a scan target was defined in OpenVAS under Configuration > Targets. A new target named "PROJECT #1 (First Vuln Scan)" was created pointing to 192.168.255.128 with the port list set to "All IANA assigned TCP", allowing simultaneous scanning via multiple IPs, with Alive Test set to "Consider Alive" to ensure the host is scanned even without a ping response.

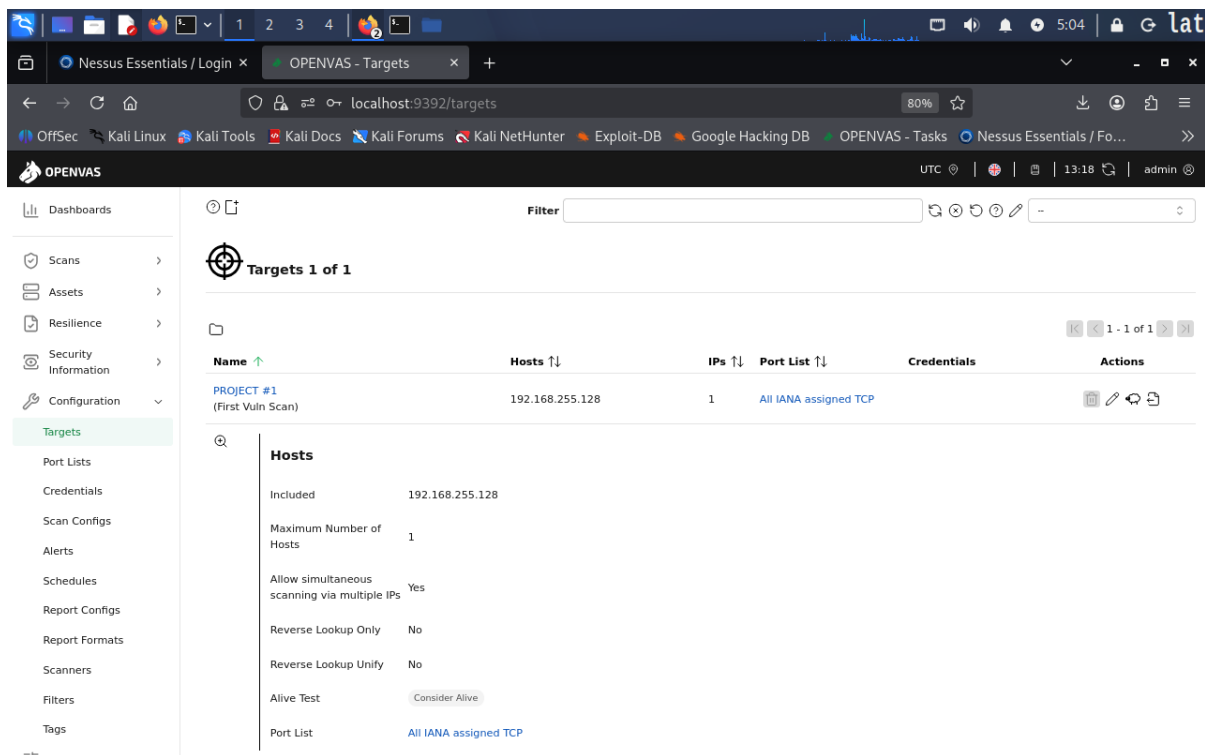


Figure 8 — OpenVAS web interface (localhost:9392/targets) — Targets 1 of 1. The target entry "PROJECT #1 (First Vuln Scan)" is listed with: Host IP 192.168.255.128, Hosts count 1, Port List: All IANA assigned TCP, Credentials: (none). Expanded host detail panel shows: Included: 192.168.255.128, Maximum Number of Hosts: 1, Allow simultaneous scanning via multiple IPs: Yes, Reverse Lookup Only: No, Reverse Lookup Unify: No, Alive Test: Consider Alive, Port List: All IANA assigned TCP.

3.3 — OpenVAS Tasks View (Scan Complete)

After the scan completed, the OpenVAS Tasks page presented three summary charts and a task list confirming the scan finished with a Critical severity rating.

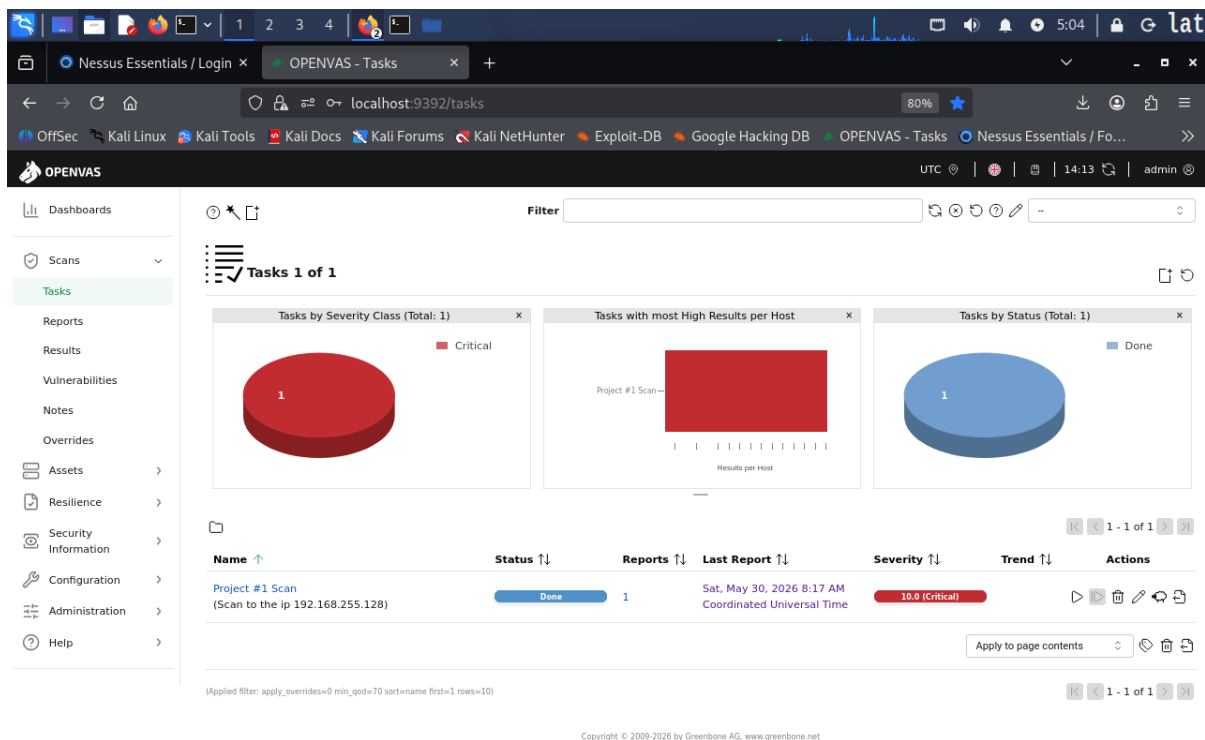


Figure 9 — OpenVAS Tasks view (localhost:9392/tasks) — Tasks 1 of 1. Three dashboard panels: (1) Tasks by Severity Class (Total: 1) — full red pie chart labelled Critical; (2) Tasks with most High Results per Host — bar chart showing "Project #1 Scan" with a large red bar; (3) Tasks by Status (Total: 1) — blue pie chart labelled Done. Task list shows: Name "Project #1 Scan (Scan to the ip 192.168.255.128)", Status: Done (blue badge), Reports: 1, Last Report: Sat May 20 2026 8:17 AM Coordinated Universal Time, Severity: 10.0 (Critical) red badge, Trend indicator present, full Actions column (play, clone, edit, delete, export, schedule).

3.4 — OpenVAS Reports View

The Reports section provided a higher-level view of all completed scans, displaying three chart panels and the report list row with a full severity breakdown.

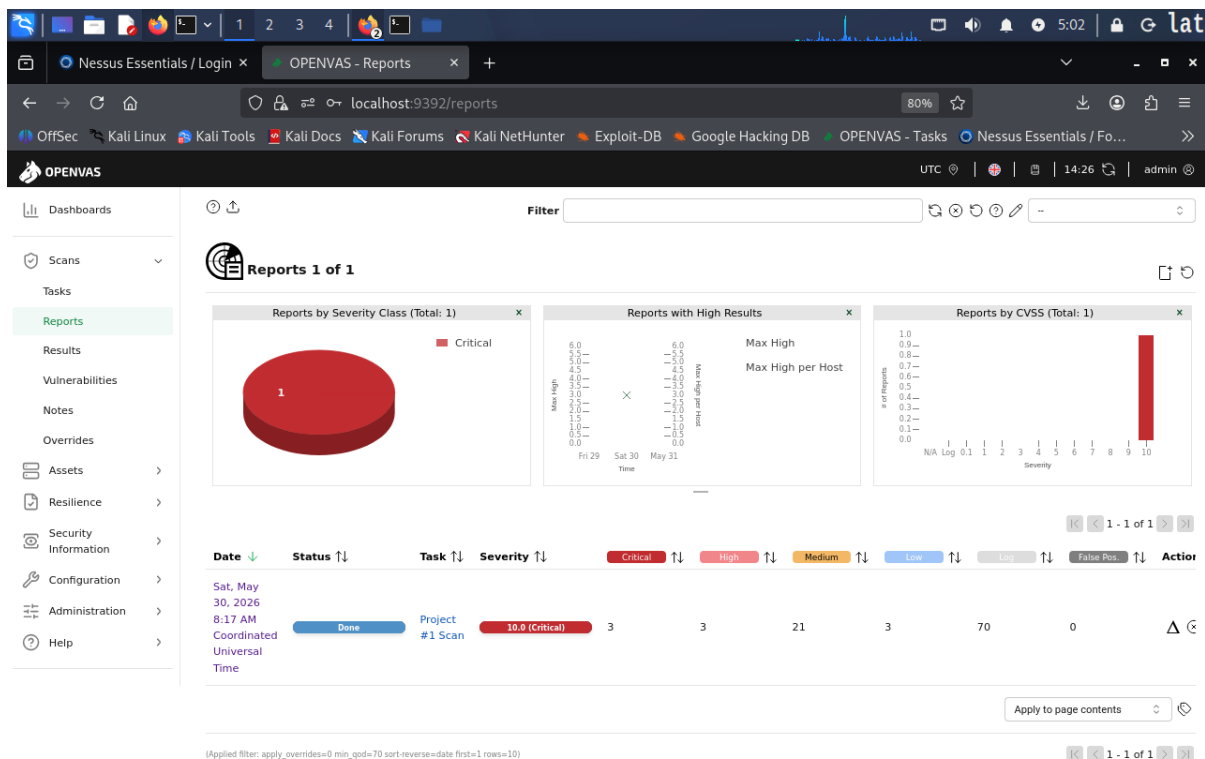


Figure 10 — OpenVAS Reports view (localhost:9392/reports) — Reports 1 of 1. Three panels: (1) Reports by Severity Class (Total: 1) — red pie labelled Critical; (2) Reports with High Results — timeline chart showing Max High and Max High per Host for May 29–31 with the spike on Sat May 30; (3) Reports by CVSS (Total: 1) — bar chart with a tall red bar at CVSS score 10 (right edge). Report list row: Date Sat May 30 2026 8:17 AM Coordinated Universal Time, Status: Done (blue), Task: Project #1 Scan, Severity: 10.0 (Critical), Critical: 3, High: 3, Medium: 21, Low: 3, Log: 70, False Positives: 0.

3.5 — OpenVAS Assets Dashboard — Most Vulnerable Hosts

The OpenVAS Assets dashboard tab aggregated host-centric vulnerability data across four widget panels, showing the target host severity profile, network topology, most vulnerable operating system CPEs, and host modification timeline.

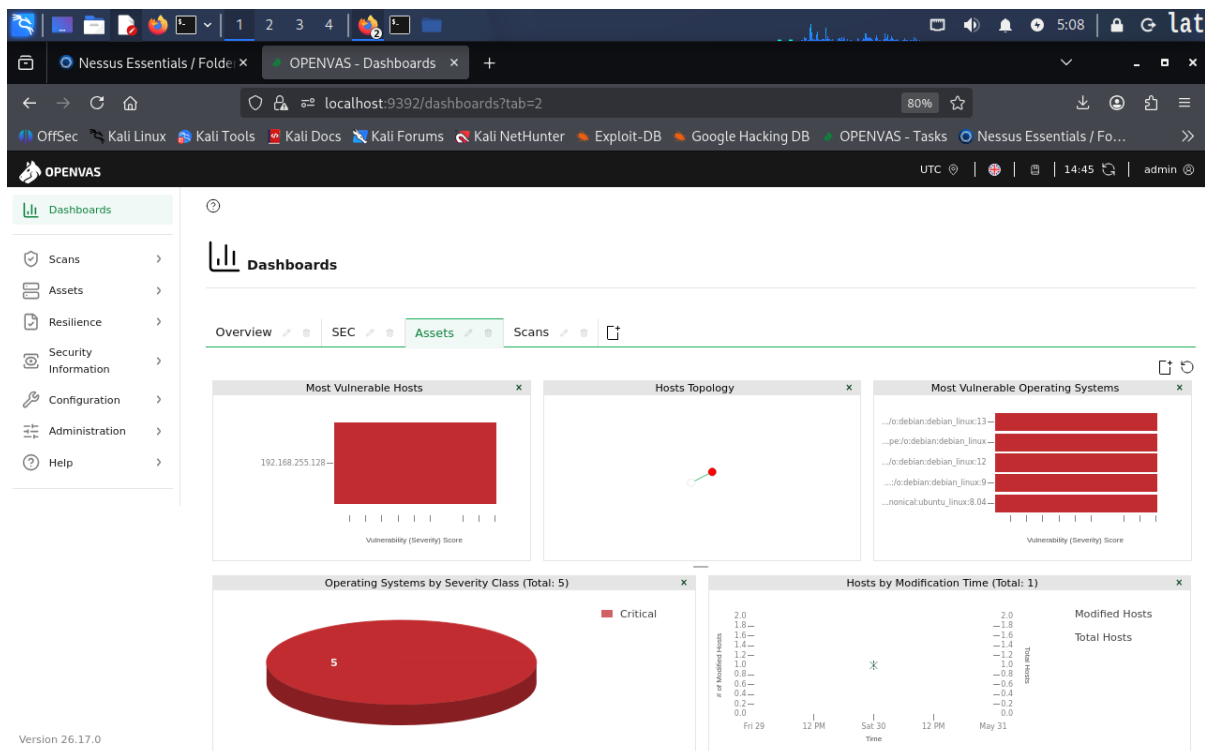


Figure 11 — OpenVAS Dashboards — Assets tab (localhost:9392/dashboards?tab=2) running GVM version 26.17.0. Four panels: (1) Most Vulnerable Hosts — horizontal bar chart with 192.168.255.128 showing a very high Vulnerability Severity Score (bar extends the full width); (2) Hosts Topology — network graph with a single red dot node representing the target; (3) Most Vulnerable Operating Systems — horizontal bar chart listing multiple Debian Linux CPE entries (debian:13, debian:12, debian:9, debian:8.04) plus ubuntu:8.04, all rated Critical; (4) Operating Systems by Severity Class (Total: 5) — solid red pie chart labelled Critical (5 OS entries); (5) Hosts by Modification Time (Total: 1) — timeline showing host was modified around Sat May 30.

3.6 — OpenVAS Scans Dashboard

The Scans dashboard tab provided a scan-centric overview with results distribution charts, task status, and historical reporting trends.

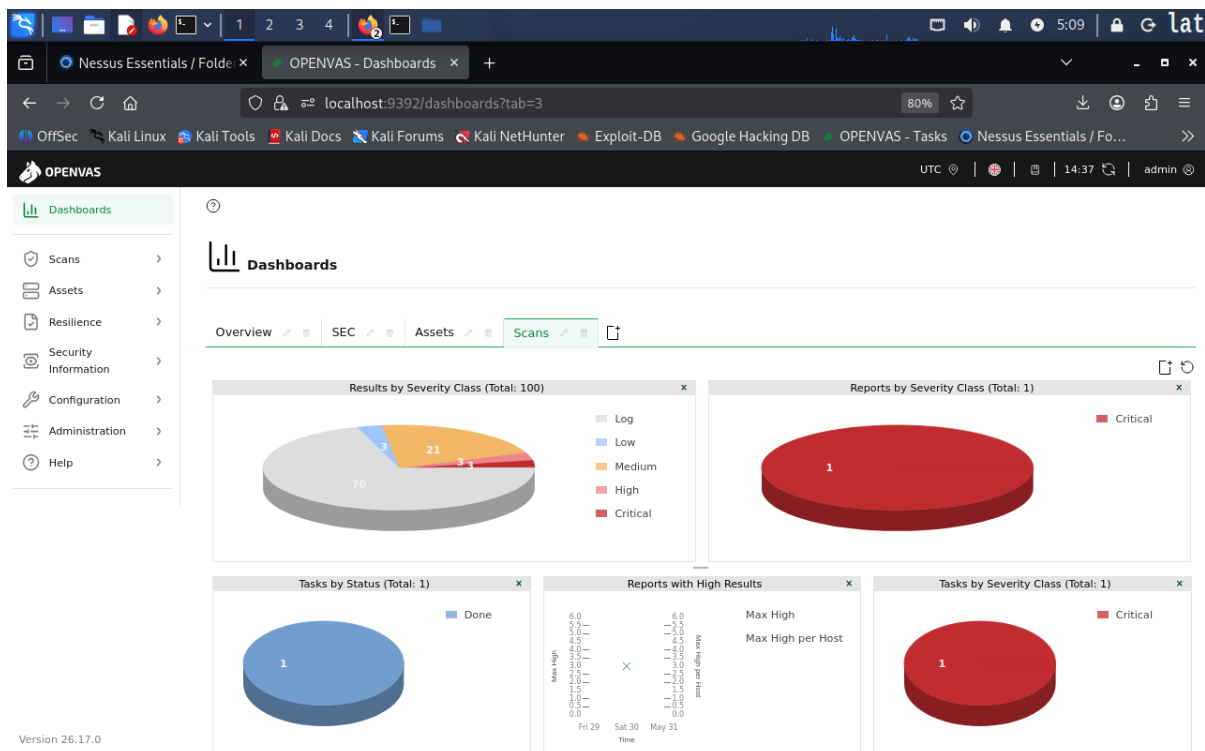


Figure 12 — OpenVAS Dashboards — Scans tab (localhost:9392/dashboards?tab=3). Six widget panels: (1) Results by Severity Class (Total: 100) — donut chart showing the distribution: Log/grey 70, Low/blue 3, Medium/orange 21, High/pink 3, Critical/red 3; (2) Reports by Severity Class (Total: 1) — red pie labelled Critical (1 report); (3) Tasks by Status (Total: 1) — blue pie labelled Done (1 task); (4) Reports with High Results — timeline for May 29–31 with Max High and Max High per Host columns; (5) Tasks by Severity Class (Total: 1) — red pie labelled Critical (1 task).

3.7 — OpenVAS Results View (100 of 441 Total Results)

The Results section listed individual scan result instances across all hosts and ports, providing granular detail beyond the vulnerability plugin summary. The page showed 100 results out of a total 441, with the top three being Critical-severity findings.

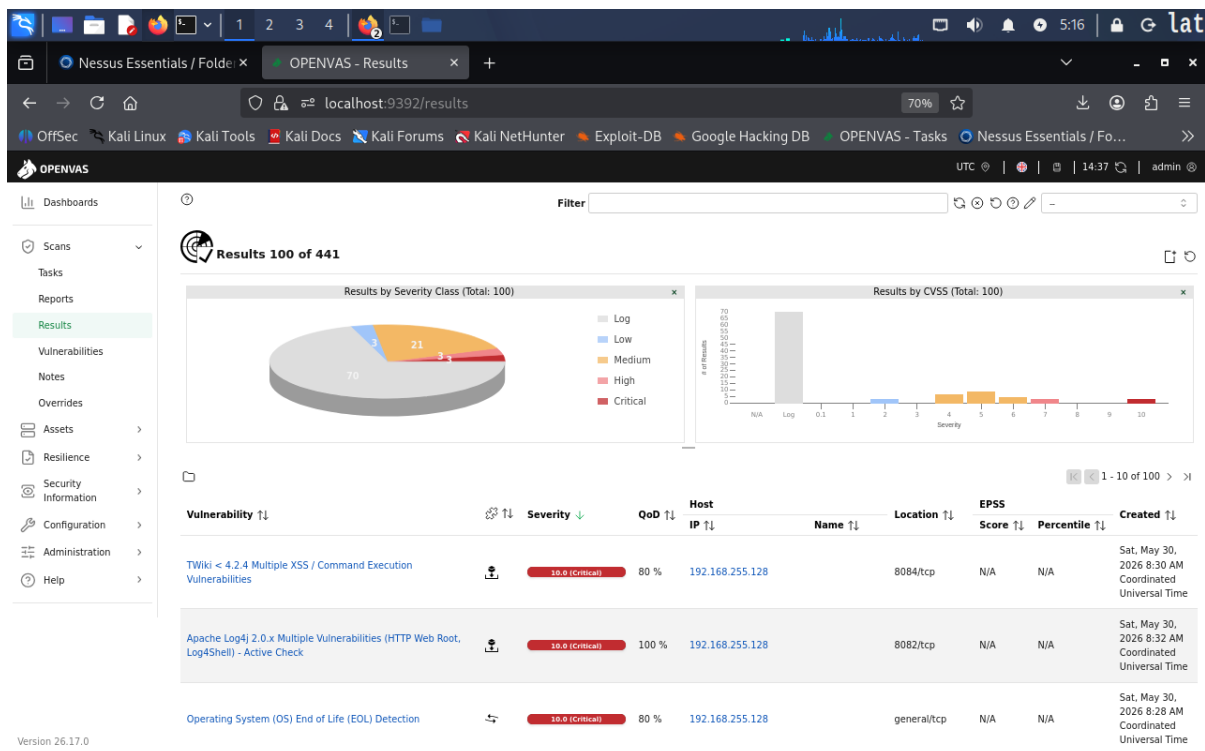


Figure 13 — OpenVAS Results view (localhost:9392/results) — Results 100 of 441. Two summary charts: (1) Results by Severity Class (Total: 100) — donut showing 70 Log (grey), 21 Medium (orange), 3 High (pink), 3 Critical (red); (2) Results by CVSS (Total: 100) — bar histogram with the majority of results clustered at lower CVSS values and Critical bars visible at CVSS 10. First-page results table: (1) TWiki < 4.2.4 Multiple XSS / Command Execution Vulnerabilities — Severity 10.0 (Critical), QoD 80%, Host 192.168.255.128, Location 8084/tcp, EPSS Score N/A, Created Sat May 30 2026 8:30 AM; (2) Apache Log4j 2.0.x Multiple Vulnerabilities (HTTP Web Root, Log4Shell) — Active Check — Severity 10.0 (Critical), QoD 100%, Location 8082/tcp, Created Sat May 30 2026 8:32 AM; (3) Operating System (OS) End of Life (EOL) Detection — Severity 10.0 (Critical), QoD 80%, Location general/tcp, Created Sat May 30 2026 8:28 AM.

3.8 — OpenVAS Vulnerabilities View (77 of 404 Unique Vulnerabilities)

The Vulnerabilities section deduplicated results into unique vulnerability entries across all hosts, revealing a much broader attack surface than the Nessus scan alone. A total of 404 unique vulnerabilities were discovered, with 77 shown on the first page.

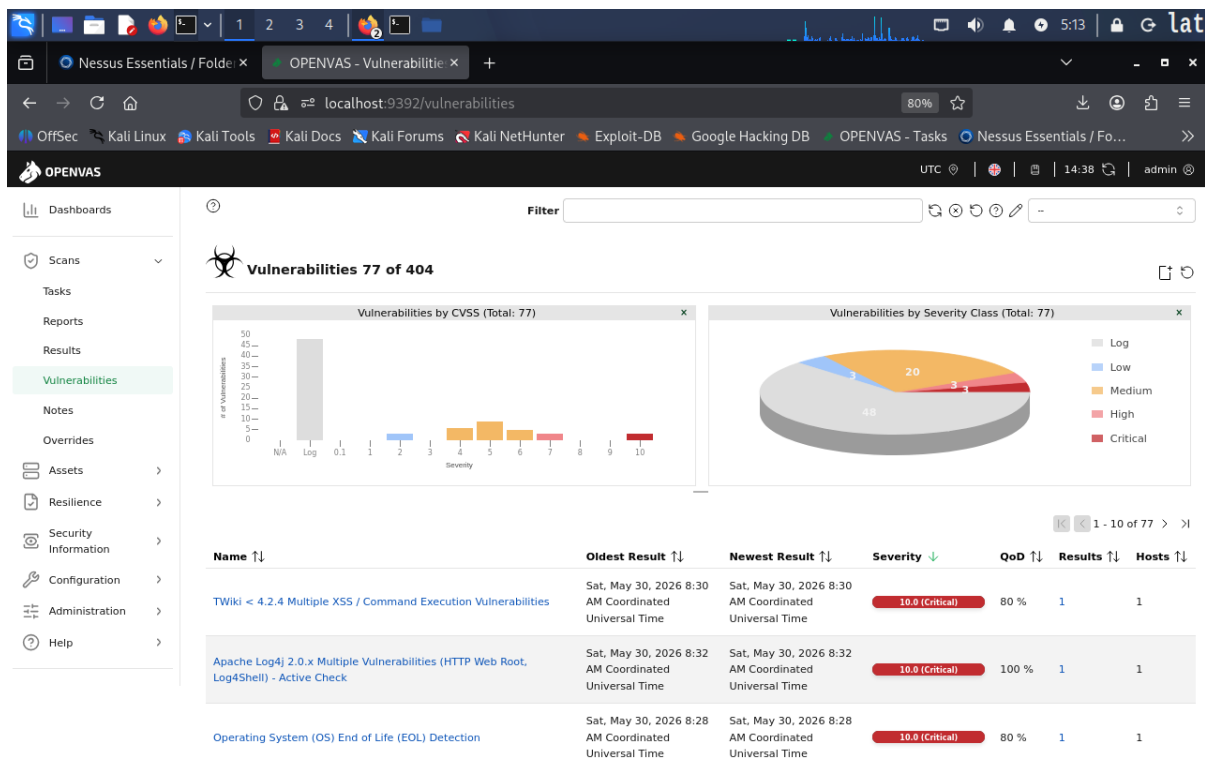


Figure 14 — OpenVAS Vulnerabilities view (localhost:9392/vulnerabilities) — Vulnerabilities 77 of 404. Two charts: (1) Vulnerabilities by CVSS (Total: 77) — bar histogram showing most vulnerabilities clustered at lower CVSS scores with notable bars at CVSS 4–7 (Medium/High) and individual red bars at CVSS 9–10; (2) Vulnerabilities by Severity Class (Total: 77) — donut chart: Log/grey 49, Low/blue 3, Medium/orange 20, High/pink 3, Critical/red 3 (labels visible). Top three vulnerabilities listed: (1) TWiki < 4.2.4 Multiple XSS / Command Execution Vulnerabilities — Oldest/Newest Result: Sat May 30 2026 8:30 AM, Severity 10.0 (Critical), QoD 80%, Results 1, Hosts 1; (2) Apache Log4j 2.0.x Multiple Vulnerabilities (HTTP Web Root, Log4Shell) — Active Check — Severity 10.0 (Critical), QoD 100%, Results 1, Hosts 1; (3) Operating System (OS) End of Life (EOL) Detection — Severity 10.0 (Critical), QoD 80%, Results 1, Hosts 1.

4. Summary of Findings

Both vulnerability scanners confirmed severe weaknesses across the Metasploitable2 target. Key findings unique to this scanning phase (beyond what Nmap already revealed) include:

- TWiki < 4.2.4 Multiple XSS / Command Execution Vulnerabilities (CVSS 10.0) on port 8084/tcp — a web-based wiki application running on the Metasploitable2 Apache DAV instance.
- Apache Log4j 2.0.x Multiple Vulnerabilities including Log4Shell (Active Check confirmed, CVSS 10.0, QoD 100%) on port 8082/tcp.
- Operating System End of Life (EOL) Detection — the base OS is confirmed end-of-life, meaning no further security patches are available from the vendor.
- OpenVAS discovered 404 unique vulnerabilities across 441 individual result instances, a far broader attack surface than the 40 Nessus plugins identified. This reflects OpenVAS's larger default scan config and NVT (Network Vulnerability Test) plugin count.
- The Nessus scan ran for 17 minutes (03:02–03:19 AM) with unauthenticated access only. An authenticated scan would expose additional internal vulnerabilities not reachable from the network.
- Both tools confirmed the target is running a Debian-based Linux OS rated Critical, with multiple EOL software components, default credentials, and unpatched CVEs covering SSH, SMB, HTTP, MySQL, and web applications.

The next phase of the lab will use these findings as input for active exploitation.